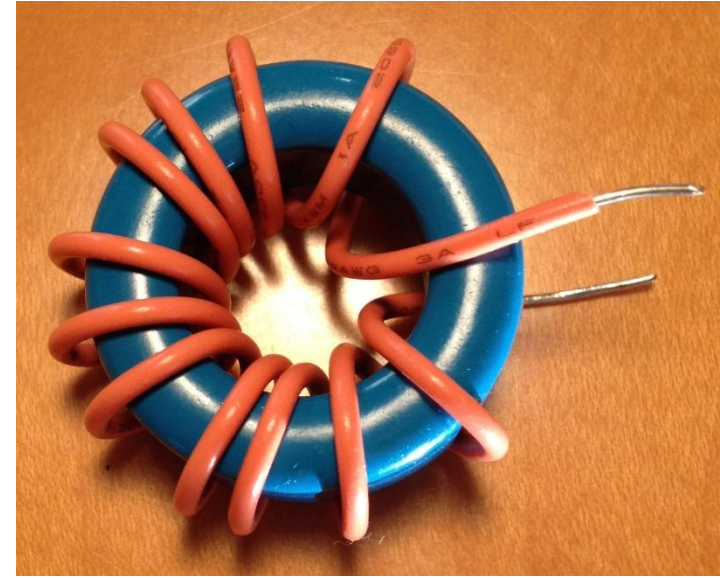


Theory behind winding inductors

- If a material is used to contain the magnetic flux:
 - Magnetic Field Strength = $\frac{NI}{\ell_e}$ (Amp Turns/meter)
 - N = number of turns, I = current, ℓ_e = effective length
 - Magnetic Flux Density = $B = \mu_r \mu_o \frac{NI}{\ell_e}$ (Teslas)
 - μ_r = relative permeability, μ_o = permeability of free space
 - Magnetic Flux = $N \mu_r \mu_o \frac{A_e}{\ell_e} I$ (Webers)
 - A_e = effective cross sectional area
 - Total Flux linked by coil = $\Phi = \underbrace{N^2 \mu_r \mu_o \frac{A_e}{\ell_e}}_{A_L} I$
 - $L = N^2 A_L$ Core manufacturer specifies A_L (Henries/Turn²)



Properties of magnetic materials

- Magnetic materials saturate. In most cases you don't want to exceed $B_{sat}/2$.
This limits the maximum current. $B = \mu_r \mu_o \frac{NI}{\ell_e}$ (Teslas)
- $\mu_o = 1.26E - 6 \frac{\text{Tesla} \cdot \text{meter}}{\text{Amp}}$

Material	μ_r	B_{sat}
Powdered Iron	~20	1T
Ferrite	~6000	0.4T
Steel	~5000	2T

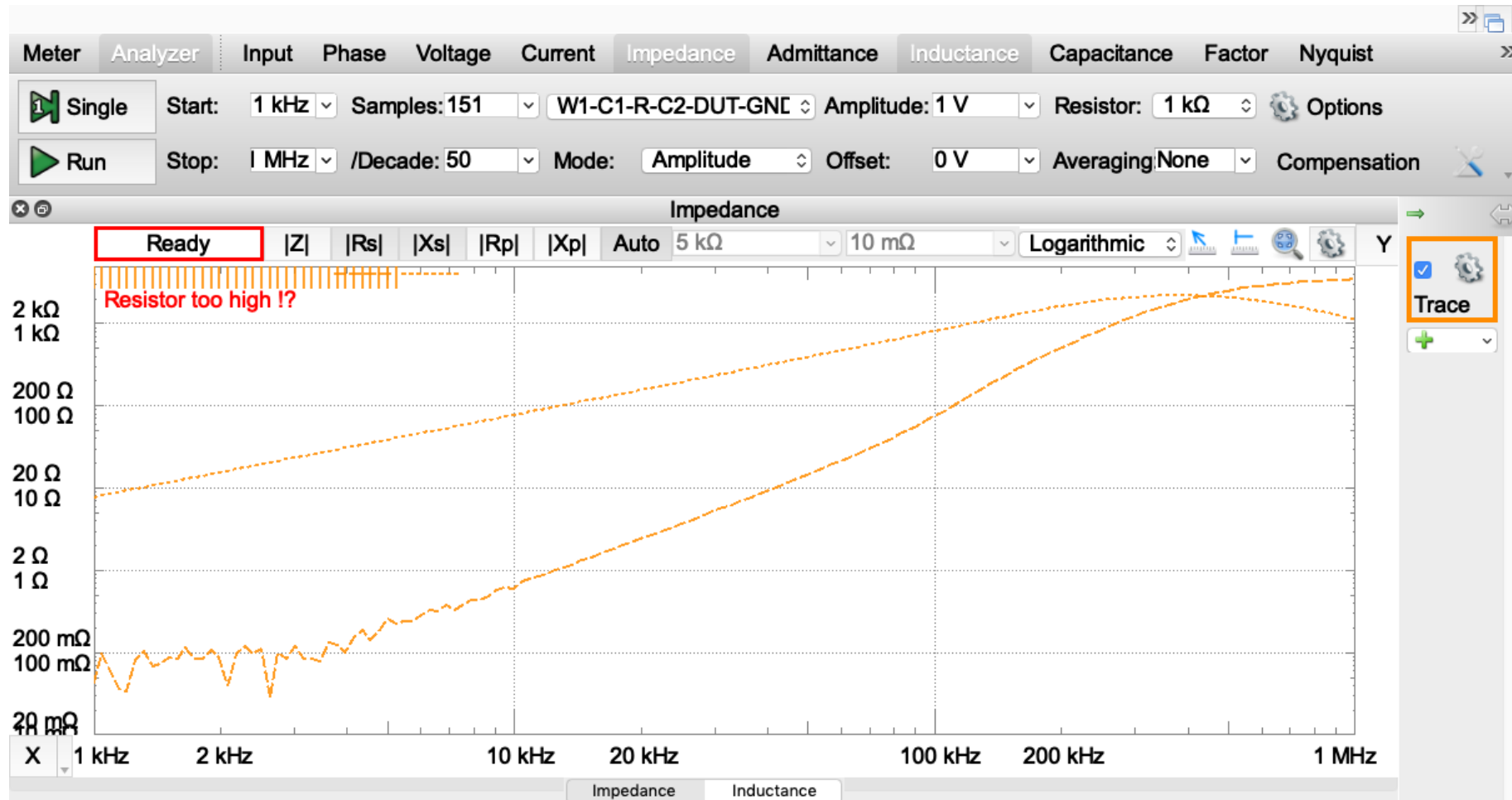
Inductor example: Make a 2mH inductor

- For our ferrite cores, the manufacturer claims $A_L = 6.9\mu\text{H}/\text{turn}^2$
- $L = N^2 A_L$
- Count a turn for every time the wire goes through the center of the toroid. Can't have fractional turns
- To figure out how much wire you need: 1.4"/turn + 3"

The AD2 enables us to automate the impedance measurement

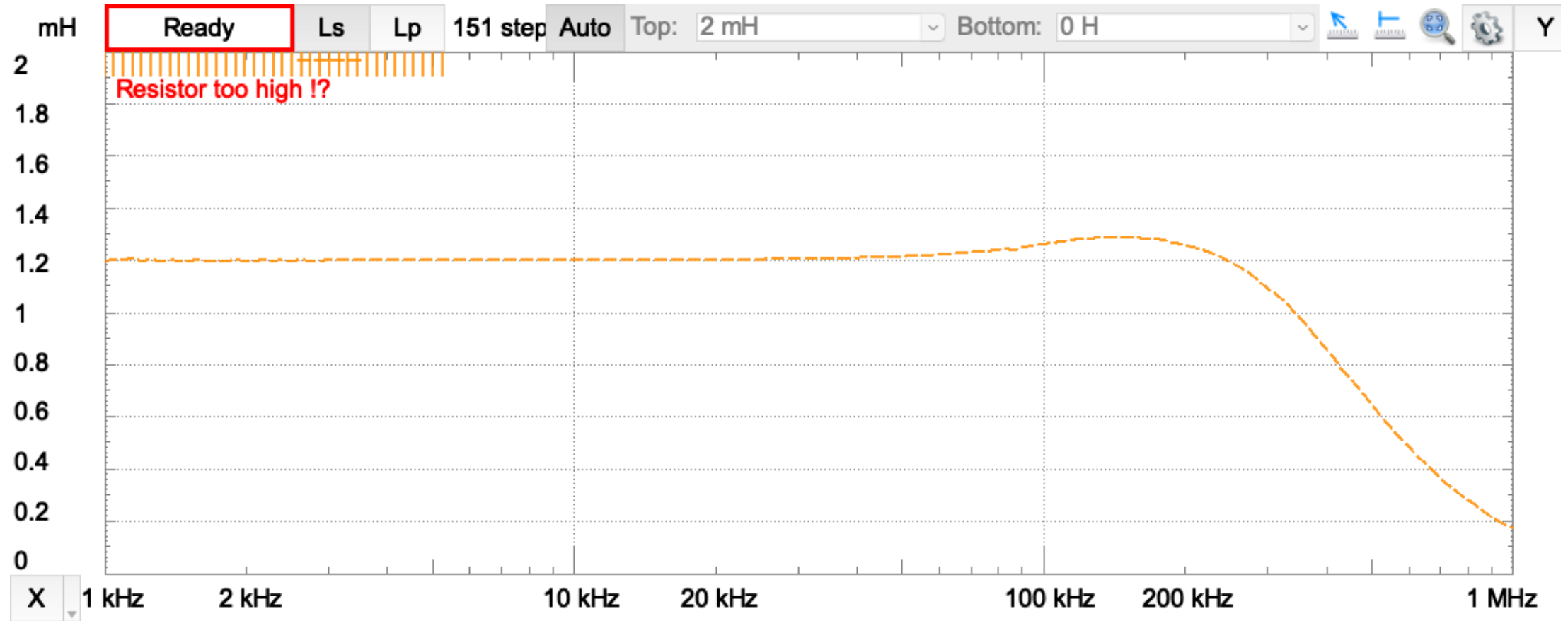
- Three configurations to choose from:
 - W1-C1-DUT-C2-R-GND
 - W1-C1-R-C2-DUT-GND
 - W1-C1P-DUT-C1N-C2-R-GND
- Must use a power of 10 resistance and specify it in the interface
- Can control the frequency range and amplitude of the input signal
- Many plotting options
 - Real and imaginary parts of impedance
 - Inductance or capacitance
 - Meter
- Watch time waveforms to make sure you aren't getting junk

Inductor example: Impedance Measurement



- Make sure to select the resistor you use in your circuit
- Note that the imaginary component increases linearly (as expected)
- The resistive component also increases with frequency, most of which is loss, not winding resistance

Inductor example: Inductance Measurement



- When plotting series inductance, the inductance should be nearly constant below 100kHz
- The nearly constant value should match our computation
- The inductance will start to fall off after about 100kHz

Inductor example: Meter Measurement

File Control View Window

Meter Analyzer Input Phase Voltage Current Impedance Admittance Inductance Capacitance Factor Nyquist Custom1 Custom2

Single ☐ Auto Frequency W1-C1-R-C2-DUT-GND Amplitude: 1 V Resistor: 100 Ω Options

Run 10 kHz Mode: Amplitude Offset: 0 V Averaging: 500 ms Compensation

Element: Auto Model: Series ☐ Custom Edit

Ls	Series Inductance	1.24 mH
Z	Impedance	77.92 Ω
Rs	Series Resistance	1.146 Ω
Xs	Series Reactance	77.91 Ω
$\angle$	Input Phase	-51.5435 $^{\circ}$
θ	Phase	89.16 $^{\circ}$
D	Dissipation	0.0147131
Q	Quality	67.96644

My Project
MyDevice
SN:210321A2A5BD
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